

Define Problem Statements

Date	14 July 2026
Team ID	XXXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	3 Marks

Problem Statement :

Conventional assessment methods primarily focus on written examinations and objective tests, which often fail to accurately evaluate a learner's conceptual understanding and verbal communication skills. There is a growing need for an AI-powered system that can analyze spoken explanations, assess conceptual clarity, and provide immediate, personalized feedback to both learners and educators.

Customer Problem Statements

Problem ID	Customer	Goal	Current Challenge	Root Cause	Impact
PS-1	A student preparing for exams or interviews	Assess conceptual understanding by explaining topics verbally.	Most assessments rely on written exams, quizzes, or multiple-choice questions rather than spoken explanations.	Traditional evaluation methods do not effectively measure verbal communication or conceptual clarity.	Students receive limited insight into their actual understanding and readiness for academic or professional evaluations.
PS-2	A teacher, trainer, or mentor	Evaluate multiple learners efficiently and consistently.	Assessing spoken explanations manually is time-consuming and difficult to scale	There are limited AI-powered tools capable of automatically analyzing conceptual understanding from speech.	Providing timely, personalized, and consistent feedback to every learner becomes challenging.
PS-3	Educational institutions and training organizations	Improve learning outcomes through intelligent assessment methods.	Existing evaluation systems emphasize written performance while overlooking verbal communication skills.	Institutions lack automated solutions that assess conceptual understanding through speech and presentation abilities.	Identifying learning gaps and making informed academic decisions becomes less effective.